GC Humanoid Robot sharing

Luke Zhang

June 2025

Robotics & Humanoid Robot Overview

Humanoid Robot is the focus application in Robotics

Industrial Robots

Heavy industrial automation

INOVANCE 汇川机器人

ESTUN ROBOTICS

- New to C3
- Mature market
- 6-axis & 4-axis Robots

- 300K units sales in 2024, -4.5% YoY
- 320K units forecast in 2025, +6.7% YoY
- Cobot share in all industrial robot: >10%

Cobots

Collaboration & automation

Specific applications: surgery

JAKA 节卡

AUBO

- Flexible production
- Labor shortage
- Fast growing

Mobile Robots

Logistic & warehouse

Hospitality Robot

Geek+

云迹科技 YUNJI TECHNOLOGY

- Logistics automation
- Last mile delivery
- Cost savings

- Huge investment to humanoid robot, forecast >10,000 units in 2025
- Various industrial & consumer scenarios for mobile robot
- GC Top 4 RVC players contribute 47% global shipment, 9.7M units

Domestic Robots

Vacuum Robots

Lawn mower robots

+ Other domestic robots

ECOVACS ROBOTICS

roborock

- Everyday helping robots
- AI drives Humanoid Robot development
- Aging population & labor shortage

Humanoid Robots

Exoskeleton

Humanoid

+ other categories

Unitree Robotics

Fourier Intelligence

Drones

Consumer

Commercial

eVTOL

dji

美团 Meituan

AUTOFLIGHT

丰翼无人机 PHOENIX WINGS

- Video & social media content creations
- Agriculture, Construction/Traffic supervision
- Delivery Drones

- Stable growth for both consumer drone & commercial drone
- EHANG EH216-S got eVTOL license

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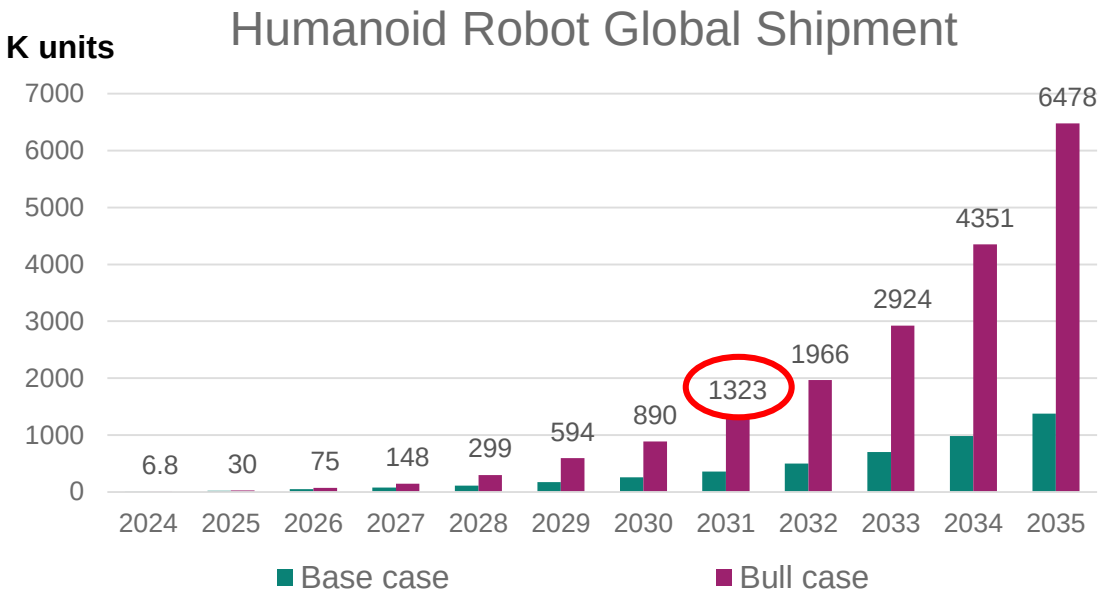
Humanoid Robot Market

Source: Goldman Sachs



2025-2035	>50%	-	Base case growth rate
CAGR	>70%	-	Bull case growth rate

Robot Unit Number (Millions) Forecast by Type Source: Citi GPS

	2024	2025	2030	2035	2050	CAGR
Autonomous Vehicle	27	34	126	380	1,858	17.4%
Domestic Cleaning Robot	286	326	541	793	1,188	5.3%
Humanoids	0	0	1	13	648	60.7%
AGV & AMR	2	3	9	28	181	17.7%
Drone	37	40	54	73	149	5.4%
Caring Robot	0	1	6	18	71	20.0%
Commercial Cleaning Robot	1	2	6	14	25	10.6%
Food & Grocery Delivery Robot	0	0	4	11	19	19.2%
Food Service Robot	0	0	3	10	15	17.2%
Caring Overlap	0	0	(1)	(5)	(18)	
Total	354	405	749	1,337	4,136	9.7%

- 1M units shipment milestone for Humanoid Robot in 2030-2031
- Humanoid Robot forecast in GC:
  - 10,000 units in 2025, ~200% YoY growth, accounting for >50% global market share
- Elon Musk's forecast for Tesla's Humanoid Robot:
  - 2-5K units in 2025; 20-50K units in 2026; 200-500K units in 2027

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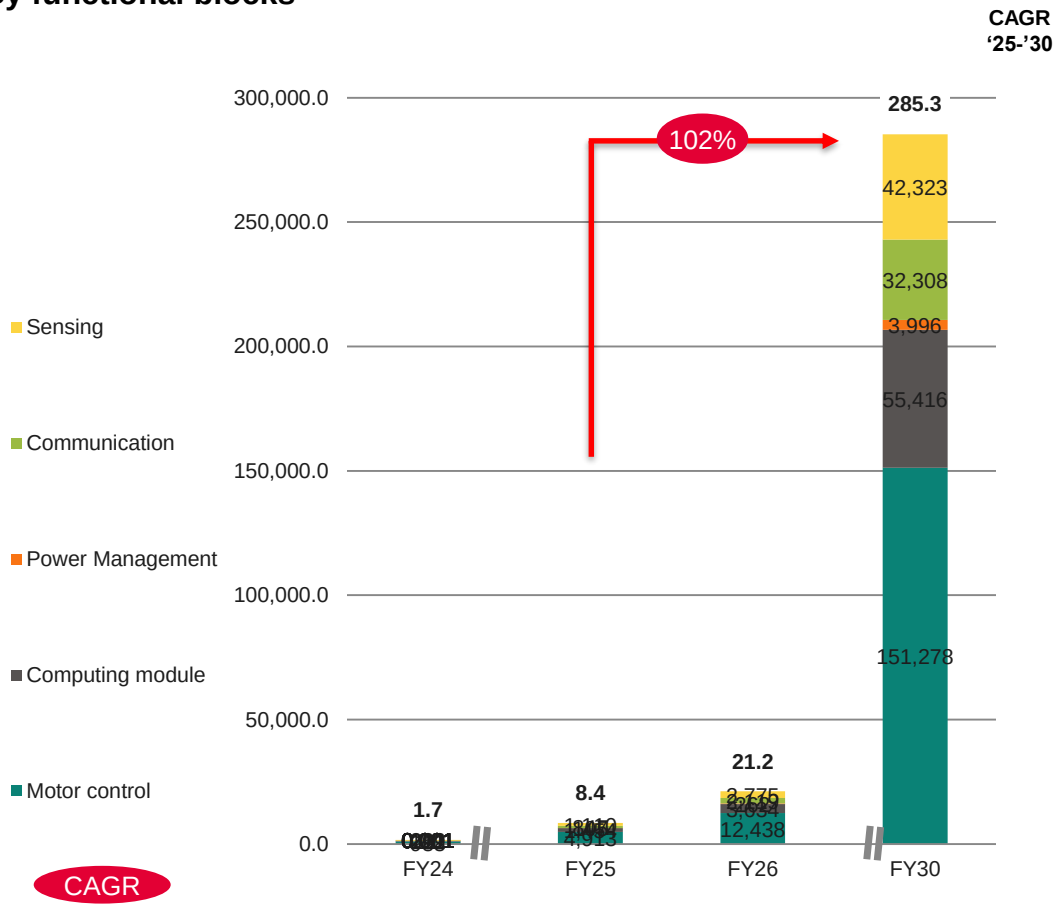
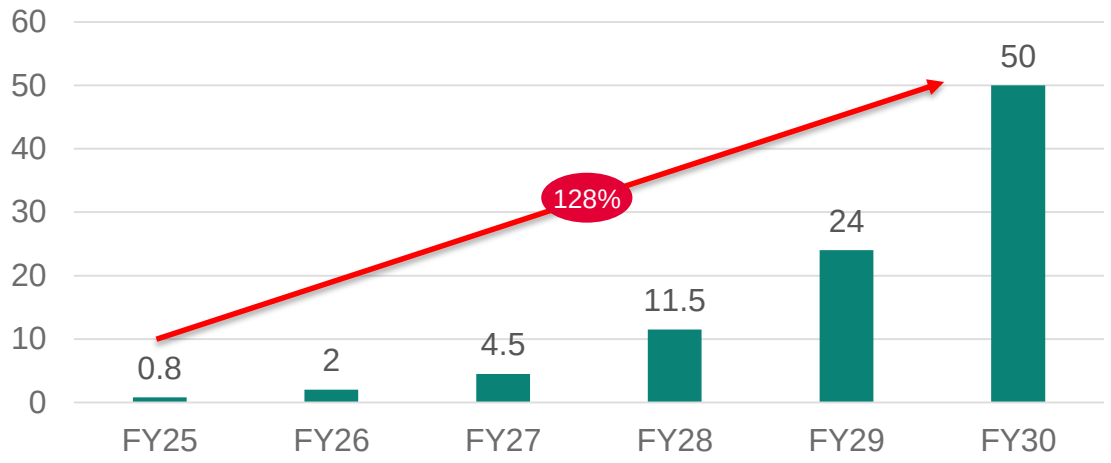
Humanoid Robot SAM & Rev Development

GC Humanod Robot market SAM (Million EUR)

By functional blocks



GC Humanod Robot Rev forecast (Million EUR)



Key Insights

- Fast evolution of AI is accelerating humanoid robot's development
- **Besides start-ups, more and more Giant companies invest resource in humanoid robot**
- Automotive manufacturing site is a popular scenario at current stage
- Potential great cost reduction for humanoid robot in following years

Source:
1) IFX internal assessment for humanoid robot market model
2) GC account for 50% global shipment

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More and more players are entering humanoid robots

资深玩家	初创企业	跨界玩家	科技 AI 企业	原生机器人厂商
优必选	傅利叶	小鹏汽车	科大讯飞	遨博智能
波士顿动力	智元机器人	特斯拉	达闼机器人	埃斯顿酷卓
乐聚机器人	宇树科技	追觅	阿里巴巴	天链机器人
钢铁侠科技	众擎机器人	小米	Meta	普渡机器人
伟景机器人	逐际动力	戴森	字节跳动	博实股份
Engineered Arts	银河通用机器人	三星	华为	埃夫特
Asimo	开普勒机器人	广汽		凯尔达
Agility Robotics	星动纪元	比亚迪		越疆机器人
	Figure AI	美的		节卡机器人

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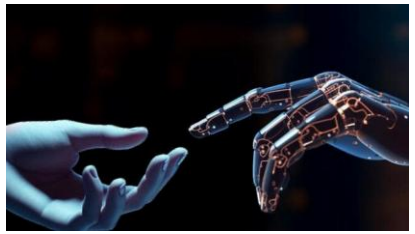
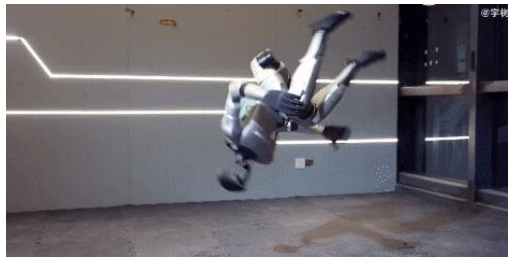
Humanoid Robot Market Overview





Market View

- Technology
  - Most humanoid robot companies focus on motion control to make robot can run/walk smoothly (acceleration by AI)
  - Fast move for new technology and innovation design in humanoid robot
  - Different understanding about humanoid robot system from customers
- Standard (Regulation)
  - No standard or regulation for humanoid robot yet, while most players believe safety will be a key for large-scale deployments
- Financing
  - Pre-IPO for Unitree & AGIBOT, target to go to public in 2026
  - ~2B Euro company valuation

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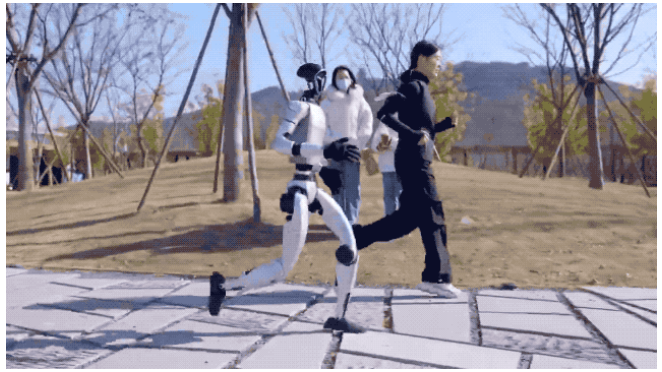
Fast evolution of GC leading humanoid robot company



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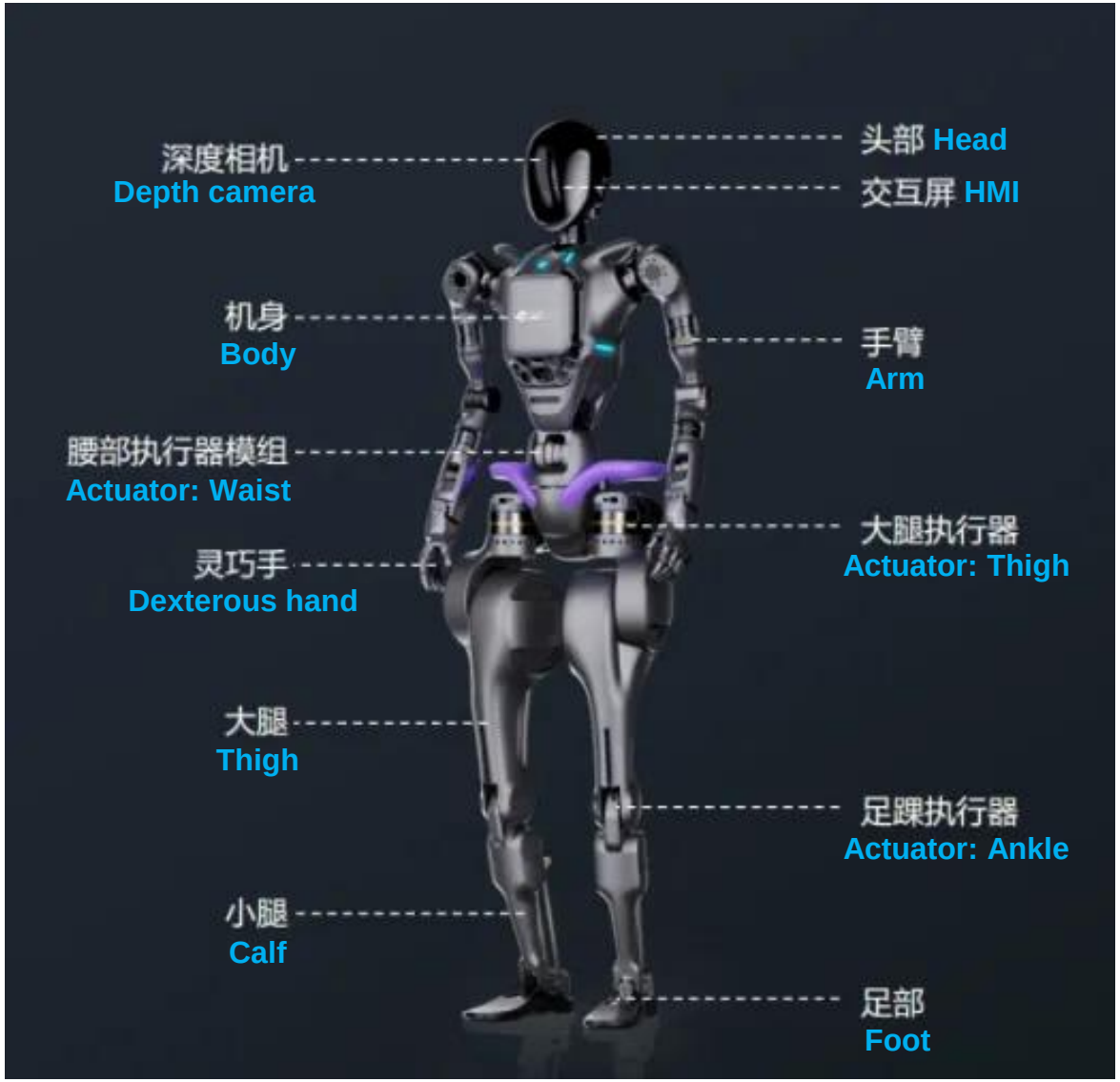
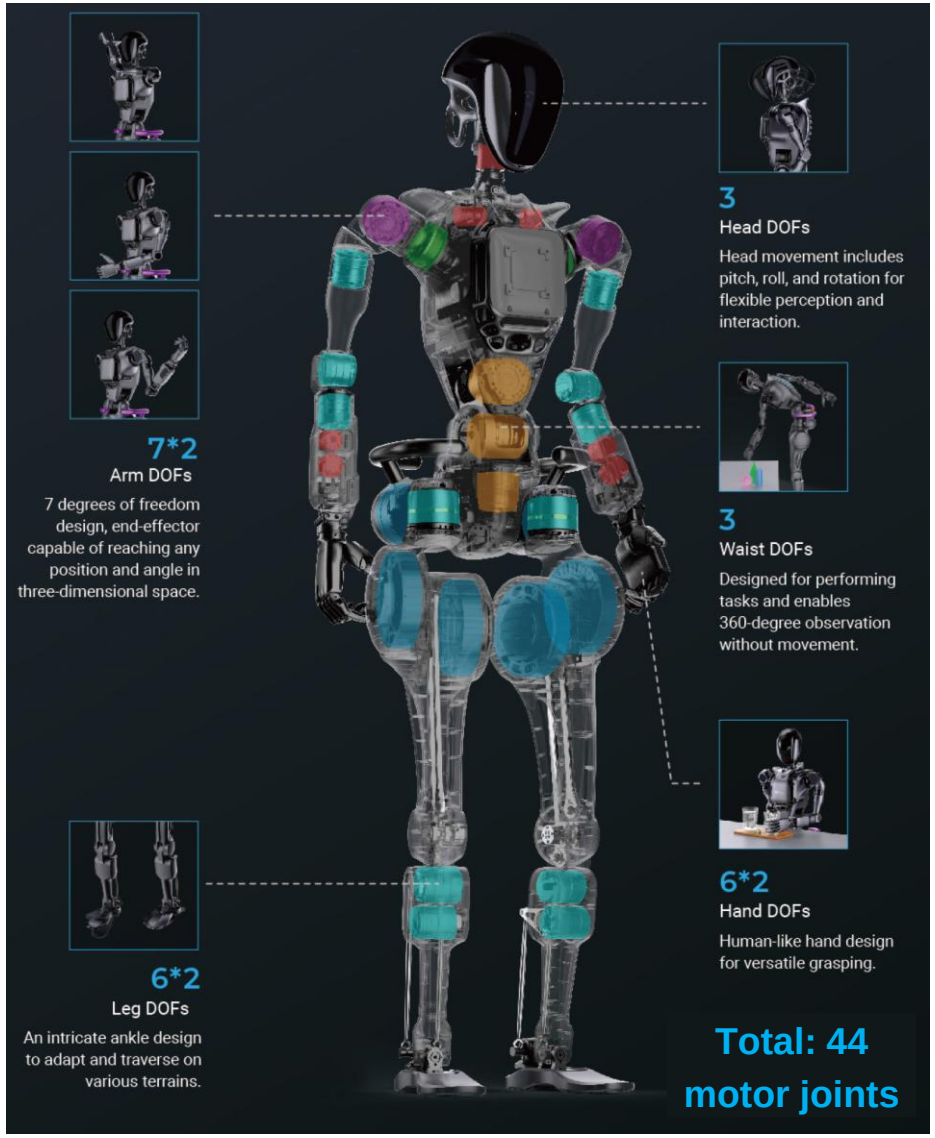
Humanoid Robot

- Opportunity
- Product & Solution Gap

Humanoid Robot structure (44 motors or even more)





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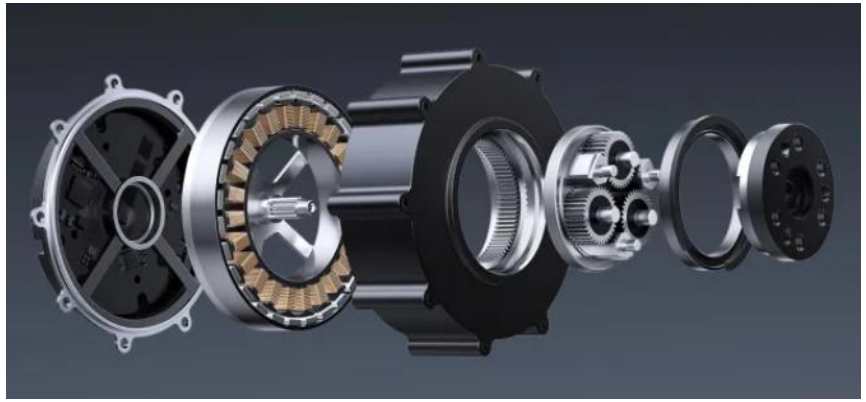
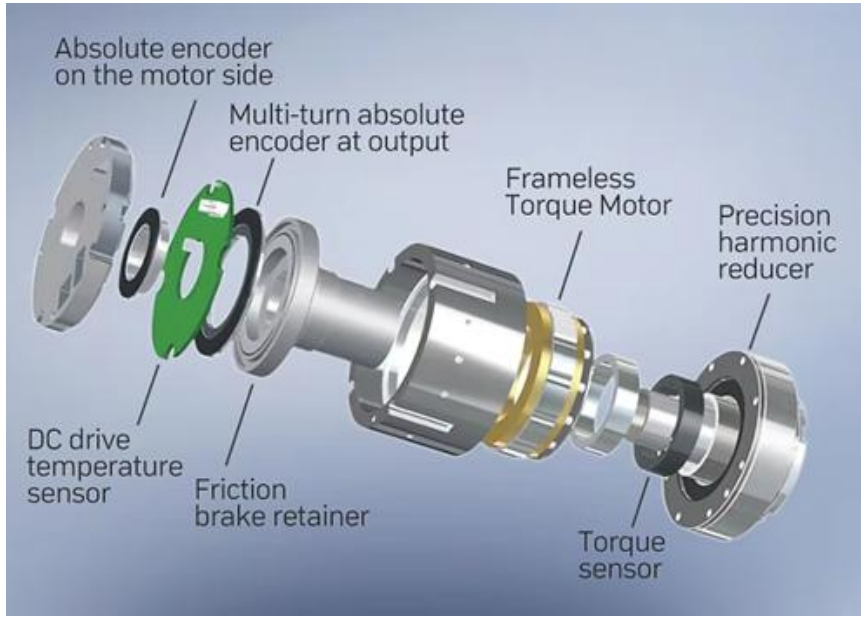
Focus functional block (1) - Joint module (motor control)





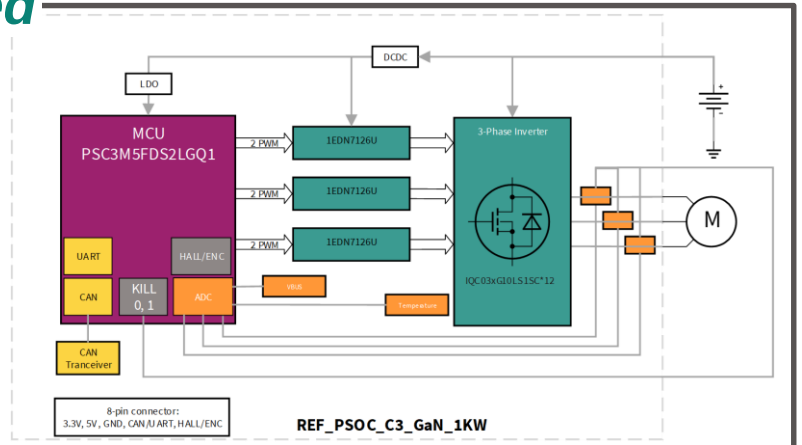
Spec & Requirement

- **Motor & Driver**: hollow shaft motor is popular with circular driver board
- **Reducer**: harmonic drive (100:1) in arm, planetary reducer (10:1) in leg
- **Encoder**: two encoders to detect motor and reducer's speed & position
- **Challenge:**
  - Limited space with many components inside
  - Peak power requirement: 3 times (rated power) & 3S
  - >30 joint modules per robot, need to control joint module's weight

Action & Support needed

Infineon total solution Demo:
(not available yet)

- MCU: PSoC C3M5 *1 (no EtherCAT)
- MV GaN: IGC033S10S1 *12
- Gate Driver: 1EDN7126U *6
- Current sensor: TLI5570/1/2 *3



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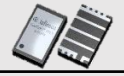
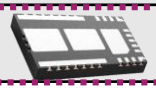
Infineon CoolGaN™ LV/MV product portfolio

Sales Code	Voltage class	$R_{DS(on),typ}$	Package type (mm)	Package figure	MP time	Positioning
IGC033S101	100 V	2.4 mΩ	PQFN 3x5 BSC		Production	p2p compatible with Navitas and package size compatible with EPC and Innoscience
IGC033S10S1			PQFN 3x5 DSC		Production	
IGB110S101		9.4 mΩ	PQFN 3x3 BSC		Production	
IGB110S10S1			PQFN 3x3 DSC		Production	
IGB070S10S1		5 mΩ	PQFN 3x3 DSC		Production	p2p compatible with standard Si products. Innoscience also recently announced plans for 100V GaN in same footprints
IGD015S10S1		1.1mΩ	RQFN 5x6 DD		Q3'25 EES Q1'25	
IGC019S06S1	60 V	1.3 mΩ	PQFN 3x5 DSC		Production	
IGC025S08S1	80 V	1.8 mΩ	PQFN 3x5 DSC		Production	
IGE033S08S1		2.3 mΩ	RQFN 3.3x3.3 SD		Q3'25 EES Q1'25	
IGC037S12S1	120 V	2.7 mΩ	PQFN 3x5 DSC		Production	
IGC090S20S1	200 V	6.7 mΩ	PQFN 3x5 DSC		Q1'25 EES available	
IGK080B041S	40 V BDS	6 mΩ	WLCSP 1.7x1.7		Q1'25 ES available	p2p compatible with Innoscience
IGK120B041S	40 V BDS	9 mΩ	WLCSP 1.2x1.7		Q2'25 EES available	
IGK048B041S	40 V BDS	4 mΩ	WLCSP 2.1x2.1		Q2'25 EES available	
IGJ024H10S2	100 V Drive	2.4 mΩ (x2)	RQFN 4.5 x 7		Q1'26 EES in Q2'25	p2p compatible with TI

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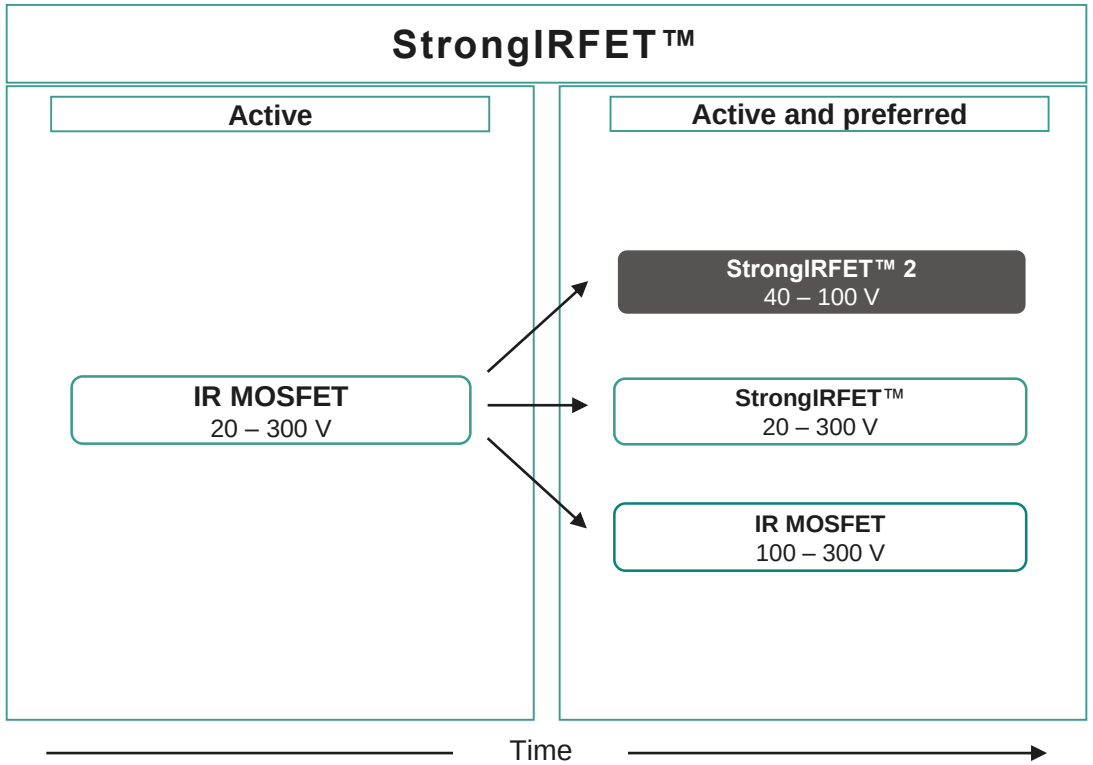
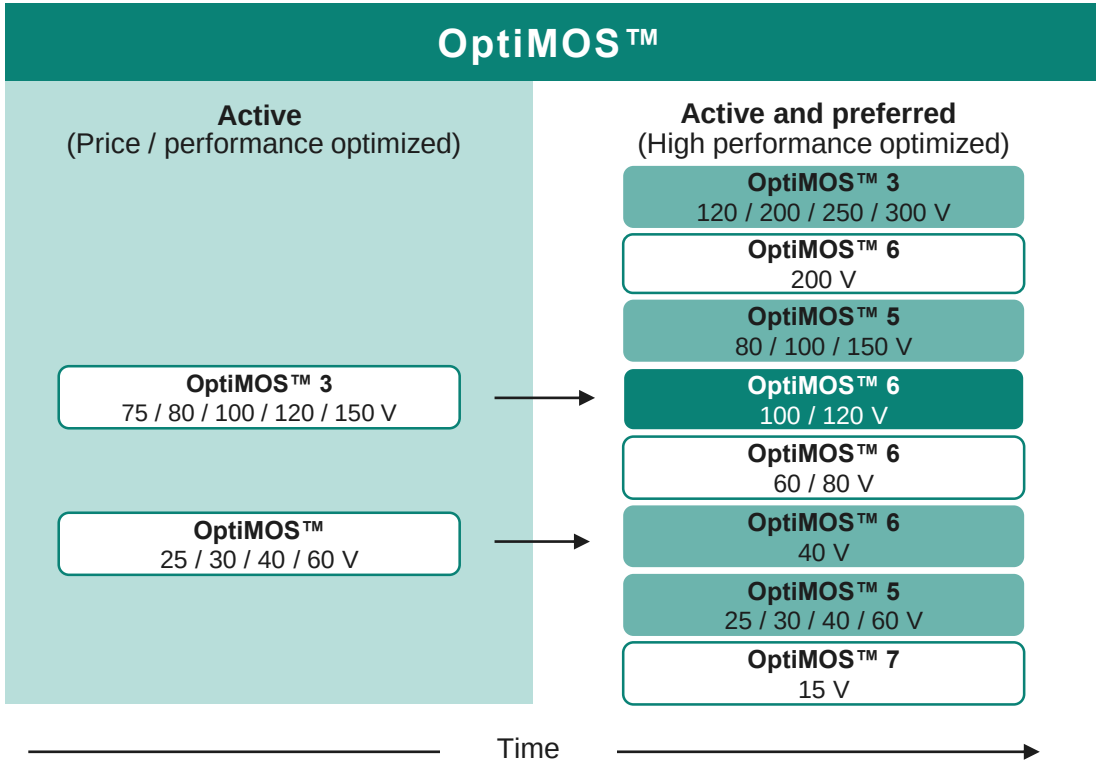
Infineon silicon advancements





StrongIRFET™

Robust and excellent price/performance ratio

Right fit products for a broad range of application

Easy of design and flexible use

OptiMOS™

Best in-class technology

Designed for high performance applications

Industry‘s best figure of merit

High efficiency, power density and current capability

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Infineon packaging advancements - 1

Industry standard package

Normal and logic-level gate drive

High current carrying capability

D^{2}PAK

10.31 x 15.88 x 4.57 mm

Industry standard package

Normal and logic-level gate drive

Up to **23%* increased current rating** vs D^{2}PAK

D^{2}PAK 7pin

10.31 x 15.88 x 4.57 mm

– Industry standard package

– **60% space reduction** vs D^{2}PAK 7pin

– Up to **63%* increased current rating** vs D^{2}PAK 7pin

TO-Leadless (TOLL)

10.10 x 11.88 x 2.40 mm

TOLG

– Optimized for TCoB performance

TOLT

– Optimized for thermal and TCoB performance

TOLG - TOLL w/ Gullwing leads

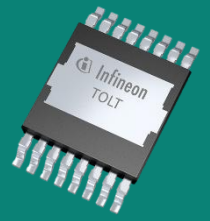
10.10 x 11.90 x 2.40 mm

– Compatible footprint to TOLL

– **Excellent solder joint reliability** on Al-IMS boards

– Up to **57%* increased current rating** vs D^{2}PAK 7pin

TOLT



10.10 x 15.20 x 2.35 mm

– **Top side cooling** package addressing **high current** applications

– Up to **52%* increased current rating** vs D^{2}PAK 7pin

* dependent on breakdown voltage of the MOSFET

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Infineon packaging advancements - 2

SuperSO8 5x6 (SSO8)

5.35 x 6.10 x 1.20 mm

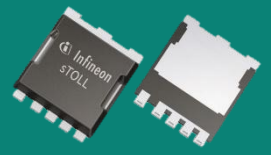
- Industry standard 5x6 mm package
- **Fused source leads** for improved current carrying capability
- High current rating

sTOLL

High current carrying capability which is ideal for surviving locked rotor events in BLDC motor applications.

Leadless package with low package resistance and minimized stray inductance.

sTOLL shrink TOLL



7.20 x 8.20 x 2.40 mm

- Bridges the gap between SSO8 and TOLL
- Infineon's solution to onsemi's 8x8 package
- **~55%*** current rating improvement vs SSO8

SuperSO8 Source-Down

Improved R_{thJA} , R_{thJC} for better transfer of power losses

Center-Gate option enables optimized parallelization

Scalable Power Block

Significantly improved form factor between 40% - 60% compared to standard SSO8 footprint

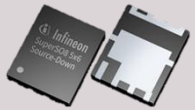
Dual-Side Cooling variant on roadmap for improved thermals

PQFN 8x6

Significantly improved form factor by ~60% compared to similarly specified TOLL device

Dual-Side Cooling variant on roadmap for improved thermals

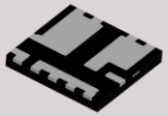
SuperSO8 Source-Down



5.05 x 6.05 x 0.75 mm

- Allows for higher Power Density in same footprint
- Same PCB routing
- **~30%*** current rating improvement vs SSO8

Scalable Power Block



6 x 6.3 mm

- High chip to package ratio for improved form factor
- Power density increased by **~50%** vs. 2xSSO8
- Simplicity in layout via half-bridge configuration

PQFN 8x6

8 x 6 mm

- Best in-class chip to package ratio
- Power density increased by **~25%** vs. PQFN 8x8
- Same pin placement as PQFN 5x6 for scalability

* dependent on breakdown voltage, $R_{DS(on)}$ of the MOSFET

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Infineon packaging advancements - 3

SuperSO8 5x6 (SSO8)

5.35 x 6.10 x 1.20 mm

- Industry standard 5x6 mm package
- **Fused source leads** for improved current carrying capability
- High current rating

SSO8 Dual-Side Cooling

- 41% volume reduction vs SSO8 for improved power density
- 95% improvement in $R_{thJC(top)}$ vs SSO8 for improved thermals

SuperSO8 Dual-Side Cooling (DSC)

5.05 x 6.05 x 0.75 mm

- Same footprint as SuperSO8 5x6
- **Exposed top** for dual-side cooling, improved thermals
- Similar current rating to SSO8

SuperSO8 Source-Down DSC

- Improved $R_{thJC\text{-}top}$ for better transfer of power losses
- Center-Gate option enables optimized parallelization

SuperSO8 SD DSC

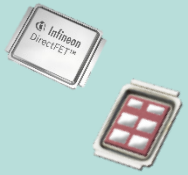
5.05 x 6.05 x 0.75 mm

- Same footprint as SuperSO8 5x6
- **Exposed top** for dual-side cooling, improved thermals
- ~30%* current rating improvement vs SSO8

DirectFET™ Mcan

- 43% volume reduction vs SSO8 for improved power density
- 90% improvement in $R_{thJC(top)}$ vs SSO8 for improved thermals
- RoHS 6/6 compliant

DirectFET™ Medium Can



5.05 x 6.35 x 0.70 mm

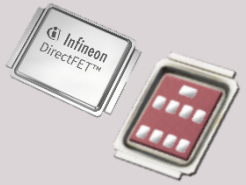
- **Source-down** for improved thermals thru the PCB
- **Exposed top** for dual-side cooling, improved thermals
- RoHS 6/6 compliant

* dependent on breakdown voltage of the MOSFET

DirectFET™ Lcan

- Limited to StrongIRFET™ silicon technology
- Can replace 2x Mcan DirectFET™, SuperSO8, SuperSO8 Super Cool devices in many apps.

DirectFET™ Large Can



7.10 x 9.15 x 0.74 mm

- **Source-down** for improved thermal thru the PCB
- **Exposed top** for dual-side cooling, improved thermals
- Similar current rating to SSO8

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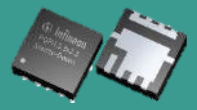
Infineon packaging advancements - 4

PQFN 3.3x3.3 SD Center-Gate

Optimized for paralleling due to center-gate pin placement and increased Creepage between source and drain

0.70 mm creepage distance vs 0.54 mm for standard Source-Down

PQFN 3.3x3.3 Source-Down (SD)



3.40 x 3.40 x 1.10 mm

Source-down for the next level of power density

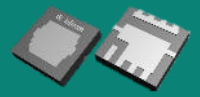
Same PCB routing

Increased fused lead area

increased I_{max}

~31%* current rating improvement vs PQFN 3.3x3.3

PQFN 3.3x3.3 SD DSC



3.40 x 3.40 x 0.75 mm

Similar benefits of PQFN 3.3x3.3 Source-Down (SD)

Increased power density and enhanced thermals due to dual-side cooling

Lower package profile

DSC = Dual-side cooling

Industry standard package

Normal and logic-level gate drive

High current rating

PQFN 3.3x3.3

3.40 x 3.40 x 1.10 mm

PQFN 3.3x3.3 Fused source leads

3.40 x 3.40 x 1.10 mm

Industry standard package

Fused source leads for improved current carrying capability

~9%* current rating improvement vs PQFN 3.3x3.3

* dependent on breakdown voltage of the MOSFET

PQFN 3.3x3.3 SD

25% improvement in $R_{DS(on)}$ and 22% reduction in R_{thJC} » increased power density

Source-down allows for improved die utilization and thermal capabilities

Large copper clip for improved current handling and thermals

PQFN 3.3x3.3 SD Center-Gate

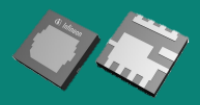
3.40 x 3.40 x 1.10 mm

Same benefits as PQFN 3.3x3.3 SD

Center-gate optimized for parallelization of MOSFETs

~31%* current rating improvement vs PQFN 3.3x3.3

PQFN 3.3x3.3 SD Center-Gate DSC



3.40 x 3.40 x 0.75 mm

Similar benefits of PQFN 3.3x3.3 SD Center-Gate

Increased power density and enhanced thermals due to dual-side cooling

Lower package profile

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Selection of Gate-driver IC's for Motor Control

Production
Development
Concept

1-channel non-isolated

1EDN-TDI Family

Truly differential inputs
4 A-8 A
SOT23-6
TSNP-6

1EDN7550B
1EDN8550B
1EDN7550U

1-channel low-side

1EDx Family

4 A-8 A
SOT23-5
SOT23-6
WSON-6

1EDN7512B
1EDN7511B
1EDN8511B
1EDN7512G
1ED44173N01B
1ED44175N01B

2-channel Half bridge/ HS+LS

IRS200X - 200 V HB & HS+LS gate driver
IRS2308 - 600 V HB gate driver

NEW

2ED2732/34/38 (HS+LS) **& 2ED2742/44/48** (HB) 160 V integrated boot strap diode, separate power and ground

6-channel

(6EDL04N02PR) /(6EDL04x06xT) - 3-phase 200 V / 600 V gate drivers with OCP, UVLO , fault and bootstrap diode

6EDL7141 - 60 V 3-phase smart gate driver

NEW

TLE9140 – 100 V 3-phase gate driver

6EDL04N03PR - 300 V 3-Phase gate driver OCP,BSD

NEW

6ED2742 - 160 V 3-phase SOI gate driver with PMU

IMD700A: XMC14xx + 6EDL7141

Production 2020 2021 2022 2023

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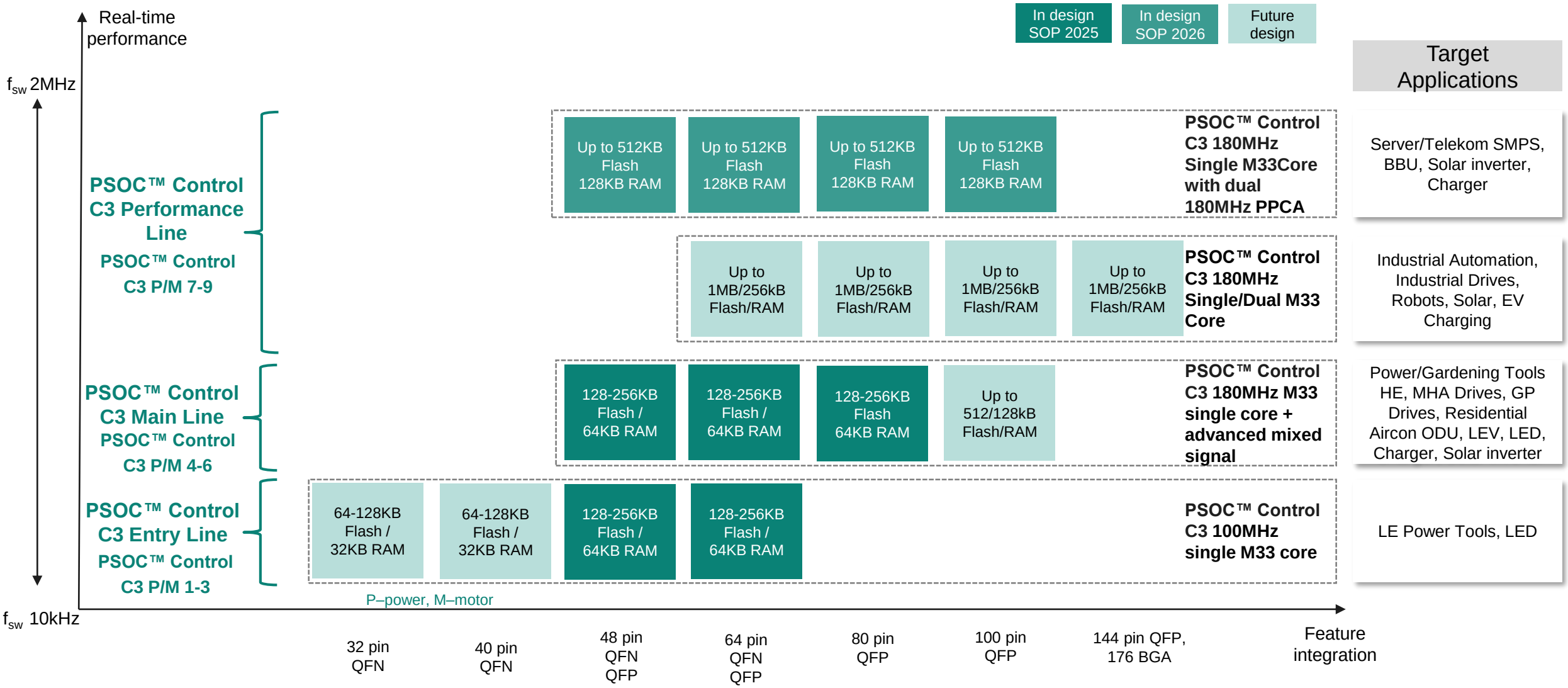
PSOC™ Control C3 Product Portfolio



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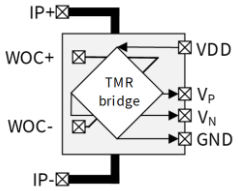
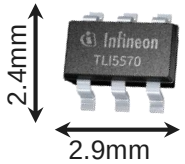
TLI557x series TMR current sensor

Preliminary data, subject to change during development

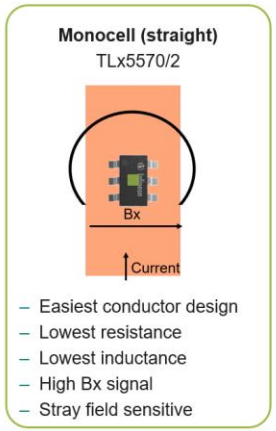
Description

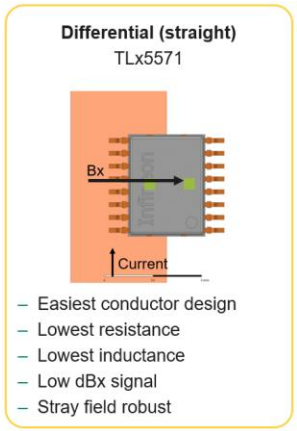
TLI5570/2 in SOT23 6pin package is a cost-competitive TMR based current sensor as Shunt replacement (non-amplified)





TLI5571 has two sensors in one chip to achieve differential detection





Features (preliminary)

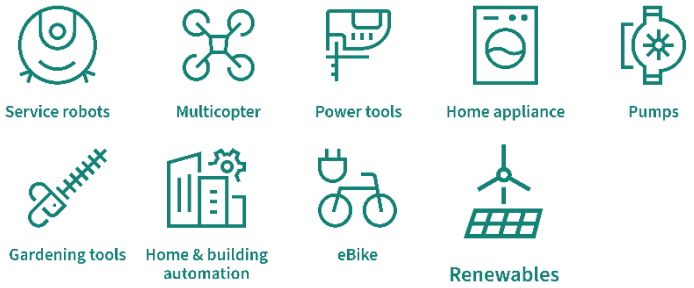
- Optimal for high currents in the range 30A - 2kA (can measure <30A, albeit with reduced accuracy)
- Miniature SMD package (2.4mm x 2.9mm)
- High bandwidth (**~500MHz**) and high accuracy (**3%-5%**)[1] & SNR (temperature trimming to be performed by the customer for achieving this accuracy)
- Discrete bridge variant with two outputs
- Contactless, galvanically isolated current measurement for reduced power losses and parasitic inductance
- Operational temperature up to **+125°C**

1) Accuracy range for temperatures -20C to 80C

Advantages & Applications

- Superior to shunt solutions due to additional features (no power loss in system, galvanic isolation, reduced BOM, smaller PCB footprint) and improved accuracy
- Cost-competitiveness on-par with existing shunt solutions
- Very few comparable products in market (from competitors)

Applications



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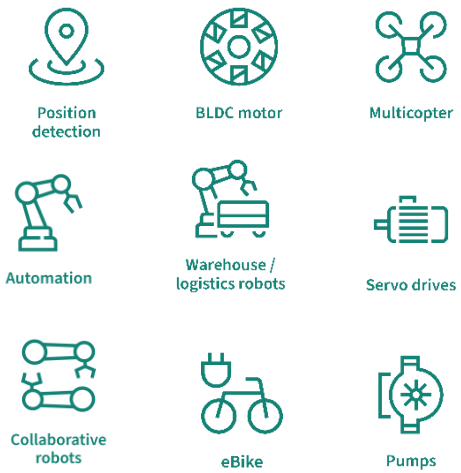
TLI49012 - Digital Angle Sensor with self-calibration feature

Target Applications

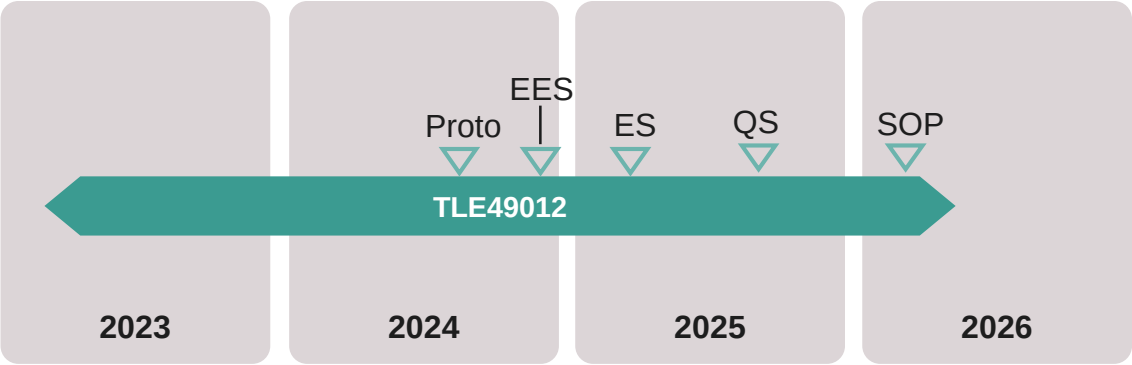
– angular position sensing applications that require high precision and digital interface



Product Highlights

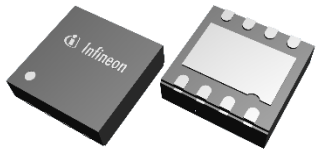
– System self-calibration provides accurate angle measurement without external calibration (<0.8°)
– 3.3V or 5V supply
– small package (3 x 3mm)
– Low latency to support reliable sensing at high dynamics (30 krpm, 25 krad/s²)
– Digital Interface: SPI, ABZ, UVW, PWM

Development Schedule



Note: Development schedule is preliminary and subject to change!

Status



VSON-8

– Product in development
– ES: available
– QS: 09/2025
– SOP: 02/2026

RoHS

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Focus functional block (2) - Dexterous Hand

Spec & Requirement

- More and more motors (6→12→16→**20+**)
- Power input: 12V, 24V, 48V(?)
- Various tactile sensor solutions while not mature enough
- **Challenge:**
  - Limited size for small motors only while need to carry heavy payload
  - Need to integrate various sensors (e.g. touch, force, temp. sensor)
  - Different robotics hands for different scenarios

Action & Support needed

- **Develop new products with high integration and small size (PL73 is evaluating the request)**
- Deep-dive sensing technology including:
  - Magnetic sensor: 3D hall sensor
  - MCU: PSoC Edge

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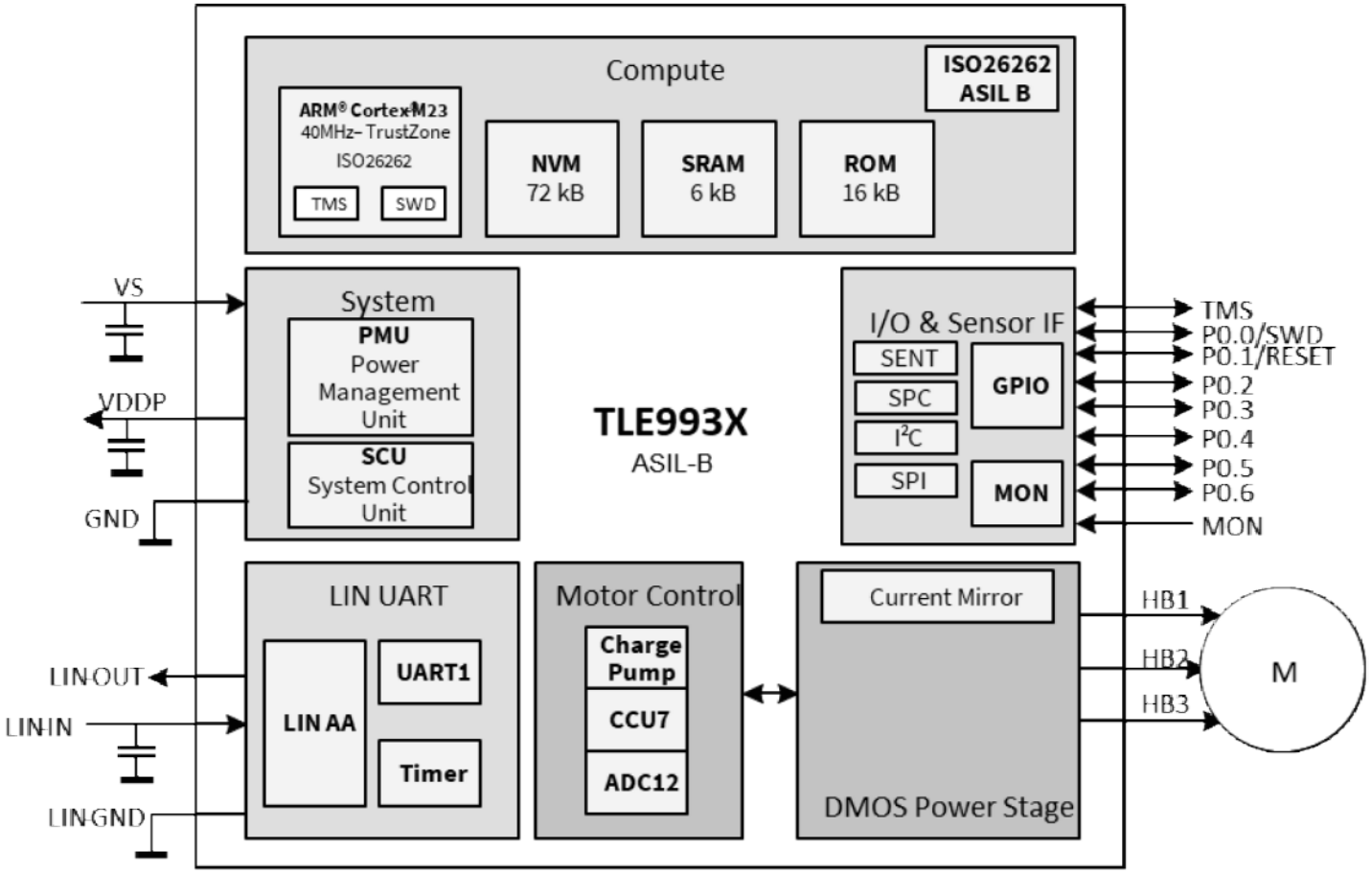
TLE993X
System-on-Chip with Arm® Cortex®-M23

Main Features

- 32-bit Arm® Cortex® -M23 core at up to 40 MHz
- up to 72 kB FLASH
- up to 6 kB RAM and emulated EEPROM
- Fully integrated B6 Bridge capable of 1A RMS / 2A Peak*
- Supply voltage range of 5.5 V to 29 V
- Integrated voltage regulator for external sensor supply
- Integrated diagnosis 12-bit ADC
- Sensor Interface (SENT, I²C, SPI, Analog Inputs)
- TSDSO-18 5x6 mm

*Depending on thermal boundry-conditions



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PSOC™ Edge E8X MCU
Enabling a New Generation of Responsive ML Edge Devices

	PSOC™ Edge E81 Base	PSOC™ Edge E83 Base + NPU + Vision	PSOC™ Edge E84 Base + NPU + Vision + GFX
Processor	Cortex-M55 + DSP Cortex-M33	Cortex-M55 + DSP Cortex-M33	Cortex-M55 + DSP Cortex-M33
Machine Learning	DSP NN accelerator, NNLite	DSP NN accelerator, NNLite, U55-128	DSP NN accelerator, NNLite, U55-128
SRAM	Up to 4 MB (System RAM) Up to 1 MB (Low-Power Domain)	Up to 4 MB (System RAM) Up to 1 MB (Low-Power Domain)	Up to 5 MB (System RAM) Up to 1 MB (Low-Power Domain)
RRAM / NVM	512 kB	512 kB	512 kB
External Memory	2x SMIF, 2x SD Host Controller	2x SMIF, 2x SD Host Controller	2x SMIF, 2x SD Host Controller
Audio/Voice	ULP Always ON prog. analog for voice, audio, sensing 4x Analog Mic, 6x Digital Mic Acoustic Activity Detection & NNLite Wake Word	ULP Always ON prog. analog for voice, audio, sensing 4x Analog Mic, 6x Digital Mic Acoustic Activity Detection & U55 ML-based Wake Word & Full Voice Inferencing	ULP Always ON prog. analog for voice, audio, sensing 4x Analog Mic, 6x Digital Mic Acoustic Activity Detection & U55 ML-based Wake Word & Full Voice Inferencing
Graphics	No	No	LP 2.5D GPU Up to 1024 x 768, MIPI-DSI/DBI formats
Vision	No	Position Detection/Face Recognition/Object Detection (VGA) via USB	Position Detection/Face Recognition/Object Detection (VGA) via USB
Peripherals & IO	USB, 10/100 Ethernet, CAN, SPI, UART, I2C, I3C, I2S	USB, 10/100 Ethernet, CAN, SPI, UART, I2C, I3C, I2S	USB, 10/100 Ethernet, CAN, SPI, UART, I2C, I3C, I2S
Security	Secured Enclave, Edge Protect Category 2 and 4	Secured Enclave, Edge Protect Category 2 and 4	Secured Enclave, Edge Protect Category 2 and 4
Packages	BGA-220, 10x10, 0.65p WLB-154, 4.3x5.3, 0.35p	BGA-220, 10x10, 0.65p WLB-154, 4.3x5.3, 0.35p	BGA-220, 10x10, 0.65p eWLB-235, 7x7, 0.4p WLB-154, 4.3x5.3, 0.35p
Availability	Samples – Q2-2025 Production – Q2-2025	Samples – Q2-2025 Production – Q2-2025	Samples – Now Production – Q2-2025

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Imagimob with ModusToolbox™ from data to edge model

Build a model

Bring your own model

OR

Buy a Ready Model (no optimization needed)

Optimize & validate your model

Develop your embedded product with IFX MCUs

Imagimob Studio

Data Acquisition → Preprocessing → Model Training

- Data Import
- Data Capture
- Data Generation
- Data Augmentation
- Data Labeling
- Preprocessor / Feature extractor selection
- Model Selection
- Model Training
- Model Validation

Imagimob Studio + ModusToolbox™

Optimization

- Hardware Optimization

ModusToolbox™

.c .a

Deployment

- Code Generation
- Target Device Test and Validation
- Field Deployment

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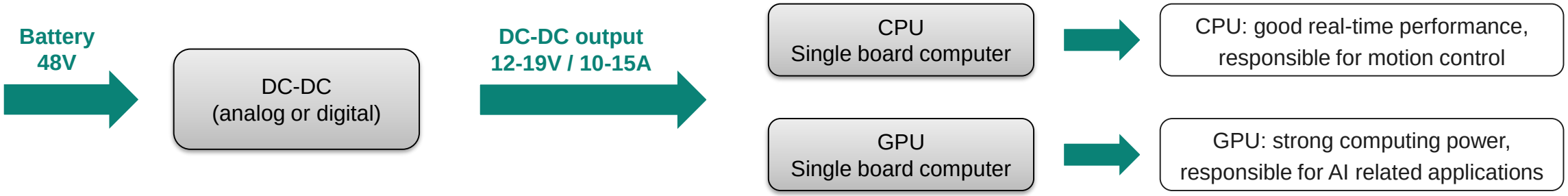
Focus functional block (3) - Power Management

Power flow for Humanoid Robot's Main Compute



Spec & Requirement

- Main Compute (CPU & GPU) boards are outsourcing and compatible with wide range input voltage (12-19V)
- Humanoid Robot customers develop the DC-DC by themselves (Buck, power for CPU, GPU and others)
- Vairous solutions in the market from different customers
- MV GaN is being evaluated as SR in DC-DC

Action & Support need

- Secure IFX share for multiphase controller & DrMOS at core board decision maker:
  - CPU: Intel
  - GPU: NVIDIA & Horizon Robotics
- Closely follow up power requirement for new NVIDIA Orin series GPU (e.g. Thor)
- **Guidance from PL88 about Shasta's competitiveness in DC-DC**

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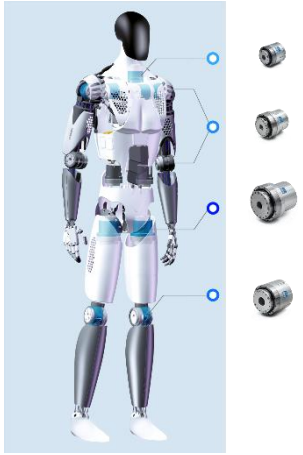
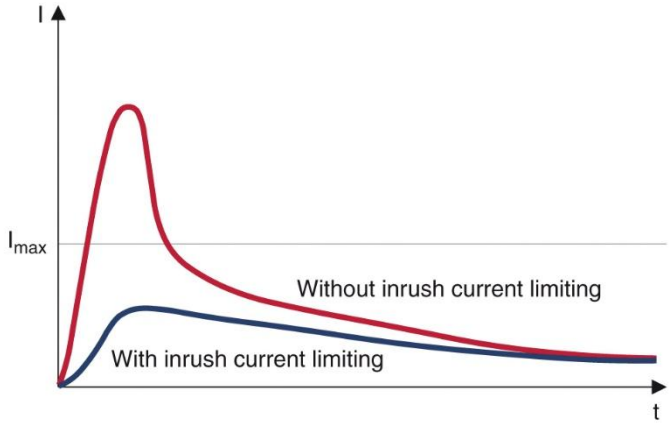
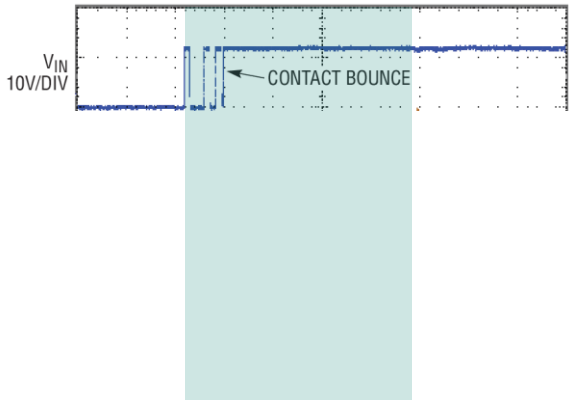
HotSwap application in Humanoid Robot

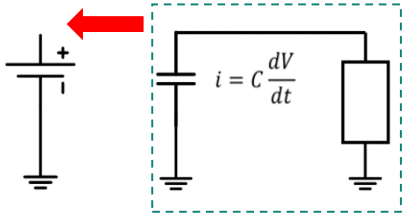
– What is Hot-Swap?



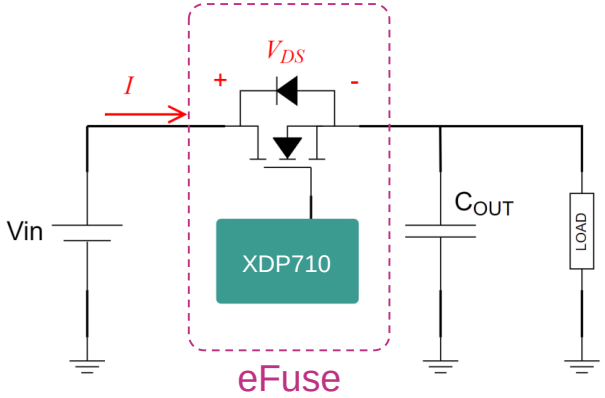


– Hot-Swap issue





– Hot-Swap controller/E-fuse



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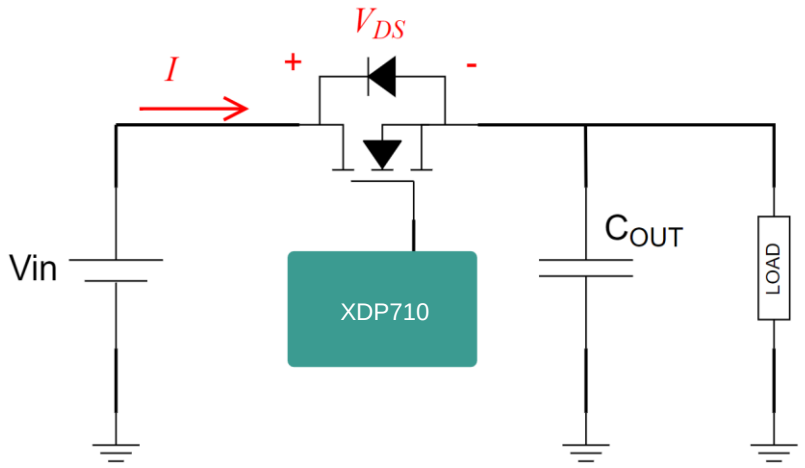
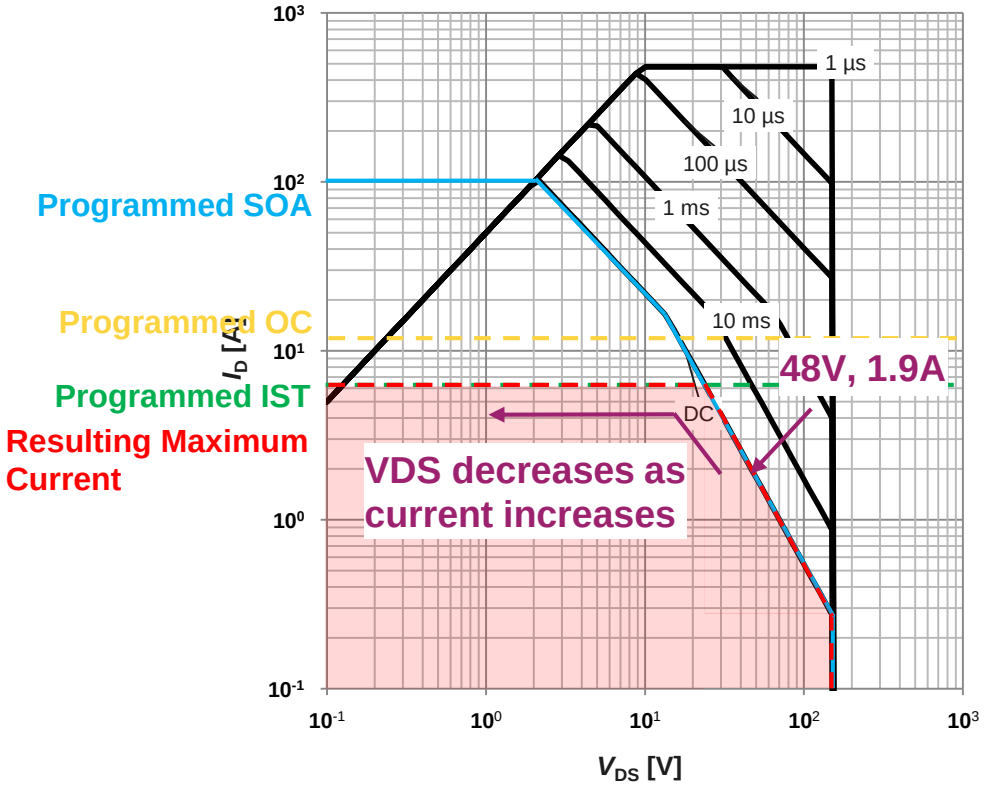
XDP710-002
Hot-swap controller for Humanoid Robot

– Our approach: “Digital“ SOA control of FET



– Advantages:
  – Safest solution – FET and Load are protected
  – Pre-programmed SOA for Infineon FETs
  – Fully optimized
  – Cost competitve
  – Fully Digital mode of operation (low cost BOM)
  – Analog-Assisted mode of operation (for backward compatability)
  – Any FET‘s SOA can be programmed using GUI



IPB072N15N3 SOA

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1EDL8011
High side Gate Driver for Humanoid Robot

Description

The 1EDL8011 gate driver IC provides **fast turn-on and turn-off** of **high side** N-channel MOSFETs with its powerful gate current capabilities. It consists of **an integrated charge pump** with an external capacitor to provide strong start-up. The gate driver IC is used to manage inrush current and protects in case of faults. Undervoltage Lockout (**UVLO**) protection at input voltage prevents the device from operating under hazardous conditions. The 1EDL8011 is available in DSO-8 package including **OCP protection** feature, an **adjustable current setting threshold**, time delay and a safe start-up mechanism with flexible blanking during MOSFET turn-on transitions.

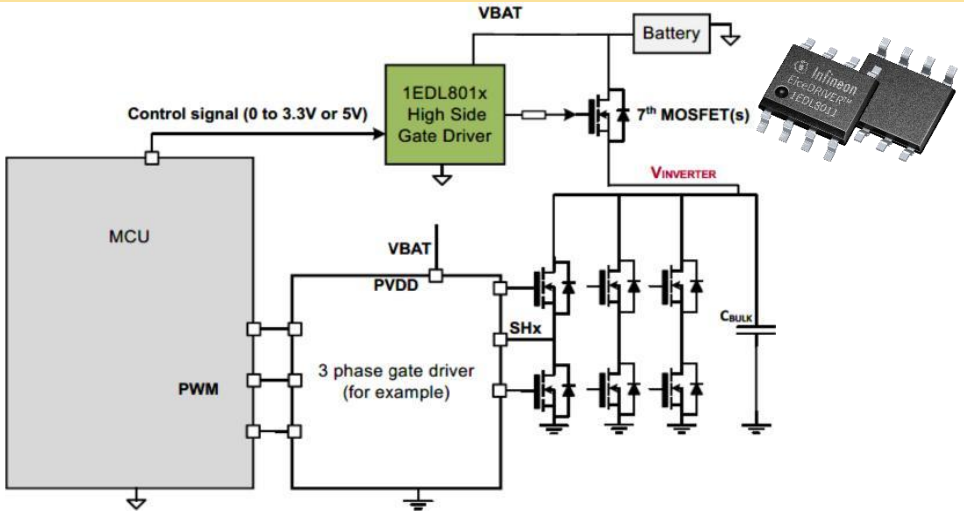
Key features

1	Integrated **125V** high side safety disconnect switch gate driver
2	Strong gate drive and charge pump for **fast turn-on/off**
3	Over-Current protection (**V_{DSON} Sense**)
4	V_{GS} safe switch off mechanism
5	**Adjustable** current trip **threshold**

Use cases and target applications

- Power tools
- Garden tools
- Robots and vacuum cleaners
- E-bikes
- Drones, multi-copters

System / application infrastructure



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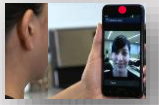
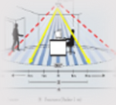
Focus functional block (4) - Environmental Sensor

Environmental	Pressure	3D Radar	3D iToF	Microphone & ultrasound TR
6x6mm² World smallest form factor	Best-in-class resolution	Application innovation leader	 Power efficient Indirect Time of Flight 3D camera	Unique SNR, sensitivity and power efficiency
CO_2 Best accuracy	Energy Efficiency	 Best performance & energy efficiency	High resolution 3D point cloud	Crystal clear audio signals

Use cases in robotics

- Collision avoidance / obstacle-stairs detection (radar,ToF,ultrasounds)
- Improved User Interface & experience (gesture, VUI)
- Height/floor positioning
- Odometry fusion / positioning / SLAM
- Vacuum flow mgt
- Material classification
- Human Presence/safety

Requires sensor+MPU/MCU+algo/software integrated system offer

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Focus functional block (5) - Connectivity (Wi-Fi & Bluetooth)

Wi-Fi 6		Wi-Fi 6E			Wi-Fi 6E SoC
2x2 MIMO	1x1 SISO	2x2 MIMO	1x1 SISO	1x1 SISO	1x1 SISO
80MHz	80MHz	80MHz	80MHz	20MHz	20MHz
Dual-band 2.4/5 GHz	Dual-band 2.4/5 GHz	Tri-band 2.4/5/6 GHz	Tri-band 2.4/5/6 GHz	Tri-band 2.4/5/6 GHz	Tri-band 2.4/5/6 GHz
Max. 1200Mbps	Max. 600Mbps	Max. 1200Mbps	Max. 600Mbps	Max. 143Mbps	Max. 143Mbps
PCIe, SDIO	PCIe, SDIO	PCIe, SDIO	PCIe, SDIO	SDIO, GSPI	SDIO, GSPI
Bluetooth 5.3	Bluetooth 5.3	Bluetooth 5.3	Bluetooth 5.3	Bluetooth 5.3(LE Audio)	Bluetooth 5.3
FCBGA, 0.65mm pitch M.2 Module WLCSP, 0.2 mm pitch SiP Module WLBGA, 0.35 mm pitch small form factor		FCBGA, 0.65mm pitch M.2 Module WLCSP, 0.2 mm pitch SiP Module WLBGA, 0.35 mm pitch small form factor		WLBGA, 0.35mm pitch Small form factor	WLBGA, 0.35mm pitch Small form factor

- **Integrated with iPA, iLNA, PMU to save BOM cost**
- **Small form factor**
- **Three products tailored to a range of applications**
- **Footprint compatible in the product family**

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System structure of Humanoid Robot

IFX offering
IFX portfolio gap
Power
Signal

Power sensor
DC-DC
48V → 12V/5V

Power AI
DC-DC
48V → 19V/10A

Battery
48V
BMS controller
MOSFET

Charger
AC-DC controller
MOSFET

Environmental sensor

- RGB Camera
- Depth Camera (iToF)
- Microphone
- Barometric sensor
- Radar
- Lidar
- Accelerometers
- Compass
- Gyroscopes
- Torque sensor
- Tactile sensor
- Temperature sensor

Decision system

GPU (NVIDIA)
AI processing

CPU (x86)
Motion control

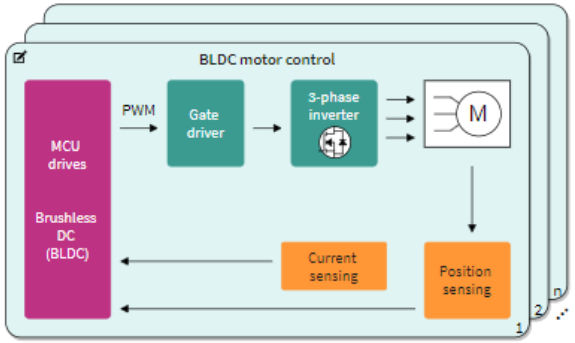
EtherCAT or CAN

Main Compute boards are outsourcing with Wi-Fi & Multiphase controller opportunities

TZTEK 天准®
AAEON® an ASUS assoc. co.

Actuator

- Arm
- Leg
- Waist / Neck
- Dexterous hand



- MCU & Gate Driver
- MOSFET & MV GaN
- Current sensor
- Position sensor

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Humanoid Robot Biz Development in GC

Secure No.1 semiconductor supplier In Humanoid Robot

Motor Control
- Secure MOSFET major share, >60%

PSoC C3 & MV GaN penetration
- DI PSoC C3, Gate Driver and MV GaN total solution at Direct Drive
- DI MV GaN at Fourier

eFuse controller
- DW XDP710-002 at AGIBOT
- DW 1EDL8011 at Fourier

Magnetic sensor
- BW and secure TLE5012B 100% share at Unitree, >200Kpcs in 2025
- DI TLI4971 at Limx Dynamics for bus current detection

Connectivity
- DW CYW20829 (Bluetooth) at Limx Dynamics
- DI CYW55572 (WiFi 6) at Fourier

Identified Top customers

Unitree Robotics
Fourier Intelligence
AGIBOT
ROBOTERA 星动纪元
逐际动力 LIMX DYNAMICS
Direct Drive Tech
ByteDance 字节跳动
mi

Success example: AGIBOT

Main Compute
NVIDIA
intel

Motor Control (Arm) via EYOUBOT
- BSC040N08NS5
- BSC030N08NS5
- 6EDL04N02PR

Power Management
- BSC117N08NS5
- BSZ075N08NS5

Motor Control (Leg)
- BSC030N08NS5
- ISC030N10NM6
- 6ED2742

eFuse
- XDP710-002
- IPT015N10N5
- IPT013N08NM5LF

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Robotics G2M Summary

Robotics G2M One Page Summary

Strategy summary

Goals & Objectives

- Target to be the market leader in focus segments

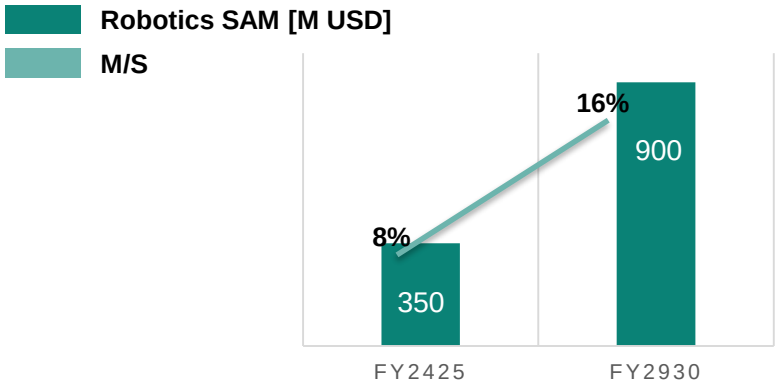
Strategy

- Win with **winners**
- Drive penetration of focus products & define **new product** requirements
- **Ecosystem** development and scale-up with **focus Disties**
- Build up local **technical competence** (application know-how)

Measures / Tactics

- **Focus products penetration at leading customers**
  - Identify & develop leading customers (Humanoid Robot: Unitree, Fourier & Limx Dynamics; Drone: DJI, Meituan & Phoenix Wings)
  - Push focus products penetration & define new products (MV GaN/PL73, PSoC C3/PL92, Angle sensor&Current sensor/PL42, Powerstage)
- **Ecosystem & Go Wider biz development**
  - Build partnership with leading system integrators, module makers, universities, alliance and core chip suppliers, e.g. D-robotics
  - Develop Focus Disty and scale the success of leading customers to C&MM market
  - Leverages Disty resource to develop Demo (e.g. Burnon/Intron PSoC C3 + GaN)
- **Local competence development**
  - Enhance IFX system technical know-how through robotics lab
  - Scale technical competence to focus Disty FAE

Opportunites



- Aging society and increased labor cost drive adoption of robotics
- The rapid evolution of AI is accelerating the development of robotics
- The concept of "embodied intelligence" (humanoid robots) was first highlighted in the 2025 Chinese Government Work Report

Risks

- IFX portfolio gap (e.g. robotics hand, 100V smart GD) to address rapid evolution of robotics
- Govt regulation/security concern might impact robotics adoption
- Uncertainty (financial and technical) with fast growing start-ups

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